Check for updates

JGR | Machine Learning and Computation

METHOD

10.1029/2025JH001021

Key Points:

- A U-Net architecture with an attention mechanism was trained on synthetic data to recognise the E, F1, and F2 layer traces in ionogram images
- The mean absolute error for automatic recognition of the layers' f_{min}, f_{max}, and h_{min} parameters, evaluated on an independent data set, ranged from 0.54 to 1.24 pixels
- The trained deep-learning models demonstrated sufficient generalisation ability to identify complex F-spread patterns

Correspondence to:

R. Sherstyukov,
Ruslan.Sherstyukov@oulu.fi

Citation:

Sherstyukov, R., Kozlovsky, A., Ulich, T., Moges, S., & Leyser, T. (2026). Synthetic-data-driven supervised learning for pixel-wise recognition of Ionogram layer traces. *Journal of Geophysical Research: Machine Learning and Computation*, *3*, e2025JH001021. https://doi.org/10.1029/2025JH001021

Received 19 SEP 2025
Accepted 15 JUN 2026

Author Contributions:

Conceptualization: Ruslan Sherstyukov, Thomas Ulich
Data curation: Alexander Kozlovsky, Thomas Ulich
Methodology: Ruslan Sherstyukov
Software: Ruslan Sherstyukov
Validation: Ruslan Sherstyukov, Alexander Kozlovsky, Samson Moges
Visualization: Ruslan Sherstyukov
Writing – original draft: Ruslan Sherstyukov
Writing – review & editing: Alexander Kozlovsky, Thomas Ulich, Samson Moges, Thomas Leyser

© 2026 The Author(s). *Journal of Geophysical Research: Machine Learning and Computation* published by Wiley Periodicals LLC on behalf of American Geophysical Union.
This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

Synthetic-Data-Driven Supervised Learning for Pixel-Wise Recognition of Ionogram Layer Traces

Ruslan Sherstyukov[1], Alexander Kozlovsky[1], Thomas Ulich[2], Samson Moges[1], and Thomas Leyser[3]

[1]Sodankylä Geophysical Observatory, University of Oulu, Sodankylä, Finland, [2]EISCAT AB, Kiruna, Sweden, [3]Swedish Institute of Space Physics, Uppsala, Sweden

Abstract The significance of automatic ionogram data interpretation is growing with the increasing number of ionosondes operating at minute-level cadence. In recent years, deep-learning models have substantially advanced ionogram pattern recognition. However, they typically require extensive labeled ionospheric data sets for training. A key challenge is developing models for pixel-wise identification of ionogram layer traces with sufficient generalization to maintain consistent performance across independent test sets from different years and phases of the solar cycle. Traditionally, entire trace labeling is not routine, as human resources are limited to determining key ionospheric parameters. We propose a method to generate ground truth without manual labeling and train a U-Net with attention for robust pixel-wise segmentation of ionogram layers at high latitudes. The model recognizes E, F1, and F2 traces and handles complex F-spread patterns and overlapping higher-order reflections. This capability was demonstrated using an independent test data set of ionograms. Following layer segmentation, we evaluated the mean absolute errors of key parameters: minimum frequency (f_{min}), maximum frequency (fmax), and minimum virtual height (h_{min}). The mean absolute errors for the E layer are f_{min} = 0.79 pixels (0.047 MHz), fmax = 0.85 pixels (0.051 MHz), and h_{min} = 0.54 pixels (3.10 km); for the F1 layer, f_{min} = 0.66 pixels (0.040 MHz), f_{max} = 0.91 pixels (0.055 MHz), and h_{min} = 0.58 pixels (3.33 km); and for the F2 layer, f_{min} = 1.19 pixels (0.071 MHz), f_{max} = 0.75 pixels (0.045 MHz), and h_{min} = 1.24 pixels (7.12 km).

Plain Language Summary Modern ionosondes can obtain data every minute, providing more information than experts can manually interpret. To address this, we implement a U-Net deep-learning model with attention mechanisms that can recognise the main ionospheric layers (*E*, F1, and F2) on ionogram images. Instead of relying on human tracings of ionospheric layers, we generated "ground truth" training data using a simple mathematical procedure that selects the strongest signal within predetermined height and frequency limits. To avoid the model overfitting and becoming too dependent on signal strength, we added random noise to input data during training. We also balanced the data set by creating synthetic F-spread rare cases. Tests with independent data set show that the model accurately identifies key parameters: minimum and maximum frequencies and minimum virtual height, with typical errors below about one pixel. This automated approach supports real-time ionospheric monitoring at high latitudes and reduces the need for expert verification, helping researchers and radio-communication system operators make faster, data-driven decisions.

1. Introduction

The study of ionospheric phenomena relies on affordable observational instruments that can provide insights into ionospheric dynamics. Instruments capable of observing ionospheric disturbances and irregularities include incoherent scatter radars (ISR), all-sky cameras, networks of GNSS receivers, and ionosondes. Each of these instruments has its limitations. ISR observations are non-continuous and spatially limited, while all-sky imagers can't operate during daytime. Over the past several decades, the most important continuous source of information on ionospheric disturbances has been dense and widespread networks of dual-frequency GNSS receivers, combined with methods for 2D and 3D ionospheric reconstruction (Norberg et al., 2018; Sherstyukov et al., 2018). Despite the high instrumental accuracy of GNSS-based trans-ionospheric sounding (approximately 0.01–0.02 TECU when measuring ionospheric phase delays), such methods may remain insensitive to certain ionospheric disturbances. This is due to the integral accumulation of phase delay and the dependence on the relative spatial arrangement between the satellite–receiver line-of-sight and the spatial structure of the ionospheric disturbance (Afraimovich et al., 2013).

29935210, 2026, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JH001021, Wiley Online Library on [06/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Ionosondes are among the most commonly used instruments for investigating the ionosphere below the height of the F2-layer electron density maximum. Their historical focus on background ionospheric plasma is largely due to the low temporal resolution of ionogram data—an approach that originated with analog ionosondes. This limitation was carried over into early digital ionosondes. However, in the past decade, many ionosondes have been configured to operate with a 1–5 minute cadence, making them capable of capturing irregular ionospheric features, particularly medium-scale phenomena such as medium-scale travelling ionospheric disturbances (Kozlovsky et al., 2025; Moges, Sherstyukov et al., 2024; Moges, Kozlovsky et al., 2024), short-lived Es layers (Kozlovsky et al., 2018; Maruyama et al., 2003; Yusupov & Akchurin, 2014) and potentially even some small-scale disturbances and irregularities with intensities dN/N < 5%–10% (Akchurin et al., 2021). A remaining challenge with ionosondes is the complexity of ionogram interpretation. The automatic scaling (i.e., determination) of ionospheric parameters, as well as the segmentation of traces—pixel-wise identification of the main traces of the E, F1, and F2 layers in ionogram images—remains challenging (see the URSI Handbook for the Interpretation of Ionograms).

ARTIST and Autoscala are widely used software tools for automatic ionogram scaling, with continuous improvements enhancing their accuracy and reliability (Galkin & Reinisch, 2008; Pezzopane & Scotto, 2005). ARTIST has evolved over several decades, introducing true-height analysis (Reinisch et al., 2005) and a confidence scoring system to assess the quality of autoscaled data, culminating in the release of ARTIST 5 (Galkin & Reinisch, 2008). Autoscala initially focused on scaling foF2 and MUF(3000)F2, and later expanded to include the F1 and sporadic E layers (Pezzopane & Scotto, 2008; Scotto & Pezzopane, 2007). A comparative evaluation of their performance using ionograms from the Rome ionospheric station showed that both programs agreed with manually scaled foF2 and MUF(3000)F2 values in approximately 95% of cases (Pezzopane & Scotto, 2007).

Deep learning architectures for image recognition have advanced significantly over the past decade, enabling their application to ionogram parameter recognition. There are two primary approaches to scaling ionospheric parameters using deep learning. The first approach involves classification and regression techniques to identify types of ionospheric layers in ionograms, followed by scaling the relevant parameters based on the classifier's output. Training these models requires ionograms as input and manually scaled ionospheric parameters as ground truth data. The extensive archive of manually scaled ionospheric parameters, created by qualified operators, provides a rich data set for training. This approach has been adopted by Sherstyukov et al. (2023, 2024) and Kvammen et al. (2024) for high-latitude regions, where the complex shapes of ionogram traces demand diverse data sets for effective model training.

The second approach focuses on identifying each pixel that corresponds to E, F1, and F2 layer traces on the ionogram. This requires ionograms as input and pixel-wise layer trace positions as ground truth data. However, such detailed ground truth data are limited and mostly collected for AI models' training. This method has primarily been applied to mid- and low-latitude regions, where ionogram trace shapes exhibit less variability than at high latitudes. It was first implemented by Mochalov and Mochalova (2019) using a U-Net architecture. Later, Xiao et al. (2020) demonstrated strong performance with a feature pyramid network, achieving average errors of 1.99 and 2.98 pixels for virtual height and critical frequency scaling, respectively. Guo and Xiong (2023) further improved results with a multi-scale attention-enhanced U-Net, reducing errors to 1.69 and 2.73 pixels. Their study utilized 19,910 ionograms collected from February–August 2018, alongside corresponding ground truth data.

Despite the strong performance of deep learning models in recognizing complete ionospheric layer traces, the limited availability of long-term, high-quality ground truth data complicates training models with sufficient generalization ability. It remains challenging to achieve consistent performance on independent test sets drawn from different years and phases of the solar cycle. Sherstyukov et al. (2024) demonstrated that combining data augmentation, filtering, and a 12-year data set reduced the root mean square error (RMSE) for the foF2 parameter on independent test data from 0.40 to 0.12 MHz compared to training on a 3-year data set. At the same time, when using a model trained on a truncated 3-year data set without augmentation, the validation errors (RMSE for foF2 = 0.27 MHz) were lower than the test errors (RMSE for foF2 = 0.40 MHz) and closer to the training errors (RMSE for foF2 = 0.17 MHz). These results indicate that model performance in ionospheric parameters recognition can be overestimated, even though validation data are not used during training, since validation sets are often sampled from temporally close periods that share similar layer trace shapes with the training data. The same performance overestimation is expected for the entire trace recognition, since ionospheric parameters critical frequencies and virtual heights of the layers are the most important part of layer trace ground truth data.

29935210, 2026, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JH001021, Wiley Online Library on [06/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

But unlike the ionospheric parameters, the long-term historical ground truth data of entire E, F1, F2 layer traces embracing at least half of solar cycle do not exist.

This challenge highlights the urgent need for approaches to train models with limited or no ground truth data while maintaining stable performance on independent data sets. Such robustness is essential for reliable automatic layer trace segmentation without operator verification. Additional motivation to train deep learning models without actual ground truth data is that transferring trained models does not always ensure consistent performance across different types of ionosondes, antenna systems, and latitudinal regions. This limitation highlights the need for methods that enable model training and adaptation to specific observational conditions without requiring significant operator resources.

The potential approach for obtaining ground-truth data of entire ionospheric layer traces involves the use of analytical ionospheric layer models, such as exponential, α-Chapman, and β-Chapman functions, which, combined with height and frequency ionospheric parameters, enable simulation of electron density profiles and generation of synthetic ionogram traces. However, these simulations would not account for the realistic quasi-oblique reflection of HF radio waves from curved electron density isolines. As a result, the ground-truth data generated using this approach will contain inherent errors.

In this study, we propose a simple approach to obtain ground-truth data of entire layer traces from ionogram images based on the argmax function, with limits derived from a data set of recognized ionospheric parameters and strict criteria to exclude incorrectly retrieved layer traces. The remaining ground-truth data satisfying these criteria, together with the corresponding ionograms, were used to train a deep-learning model for entire ionospheric layer trace recognition.

2. Method for Data Collection

In the present study, data from the Sodankylä Geophysical Observatory (SGO) ionosonde, which has been operating with a one-minute cadence since 2006, are used. For 17 years, ionogram parameters were manually scaled by a qualified operator every hour. Deep learning models for automatic ionogram parameter scaling were developed by Sherstyukov. (2025). These models were tested on an independent data set from 2021 to evaluate differences between operator and automatic scaling. The RMSE were estimated as follows: foF2 = 0.12 MHz (2 pixels), foF1 = 0.07 MHz (1.16 pixels), foE = 0.15 MHz (2.5 pixels), foEs = 0.33 MHz (5.5 pixels), $f_{\min}$ = 0.15 MHz (2.5 pixels), fbEs = 0.17 MHz (2.83 pixels), h′F = 7.7 km (1.34 pixels), h′E = 7.0 km (1.22 pixels), and h'Es = 7.1 km (1.24 pixels).These models enable the creation of a comprehensive 17-year database of ionogram parameters with one-minute time resolution. This high-temporal-resolution data set of ionospheric parameters and corresponding ionograms will serve as the foundation for generating ground truth data of E, F1, and F2 layer traces for segmentation tasks. Leveraging ionospheric parameters produced by a validated and accurate deep learning model can significantly increase the size and diversity of ground truth data of entire layer traces, which are generated using ionospheric parameters.

For the extraction of ground-truth data from ionograms, we used ionospheric frequency and virtual height parameters as bounds, which narrow the potential region of the ionospheric trace on the ionogram and reduce the risk of including irrelevant signatures, such as incorrect layer types, ground-reflected traces, X-mode traces, oblique reflections, and noise. The bounds defined by the ionospheric parameters were applied as follows:

E layer – $f_{\min}$ to foE, h′E to h′F;
F1 layer – fbEs (foE if Es absent, $f_{\min}$ if E and Es absent) to foF1, h′F to hlim (last ionogram height);
F2 layer without F1 – fbEs (foE if Es absent, $f_{\min}$ if E and Es absent) to foF2, h′F to hlim;
F2 layer with F1 – foF1 to foF2, h′F to hlim.

A deep learning model was used to estimate the layer types on the ionogram (Sherstyukov et al., 2024). To scale the h′(f) values of a particular layer, the following approach was used:

$$h(f) = \underset{h}{\operatorname{argmax}}\, A(f,h) \text{ for } f \in \left(f_{\min}, f_{\max}\right) \quad (1)$$

Despite the high accuracy of the bounds, this method is too simplistic to capture most ionogram trace patterns. To avoid selecting incorrectly segmented ionogram layer trace pixels, a strict criterion was used:

$$\left|\frac{\Delta h}{\Delta f}\right| > \text{TraceWidth}, \quad \text{for } f \in (f_{\min}, f_{\max})$$

$$\text{if } \frac{\Delta h_i}{\Delta f_i} > \text{TraceWidth}, \text{ then } \frac{\Delta h}{\Delta f} > 0, \text{for } f \in (f_i, f_{\max}) \tag{2}$$

Irregular electron-density structures—such as small-scale plasma irregularities, MSTIDs, and spread-F—create multiple ray paths with slightly different group delays. When the received chirped signal is mixed with the local reference signal, the resulting beat frequency corresponds to the signal transmission–reception time delay. Variations in group delay across multiple propagation paths broaden the beat-frequency spectrum, which manifests as the width of the layer trace on the ionogram (for a detailed description of chirp ionosonde operation, see Ivanov et al., 2003). A typical layer-trace width is estimated to be about 3–5 pixels (17–28 km). The upper range of five pixels was used in Equation 2 (for more information on the SGO ionosonde, see Kozlovsky et al., 2013).

The criterion on h(f) accounts for the potential deviation of Δh/Δf within the trace width. The first condition considers virtual height deviations during the selection of the quasi-linear part of the ionogram trace, where the electron density gradient is not steep. The second condition addresses the non-linear part of the ionogram trace, where positive dh/df may become significant due to steep electron density gradients and the accumulation of radio wave group delay near the F2 maximum height (hmF2).

Figure 1a shows ionogram layer traces scaled using Equation 1 that do not satisfy the criteria in Equation 2—the values of Δh/Δf significantly exceed the trace width (see the white solid line in Figure 1a). In contrast, Figures 1b and 1c show examples that meet the criteria. We used a data set of ionograms from 2016 to 2024, selecting one ionogram every 20 min to reduce the repetition of similar patterns. The number of cases satisfying the criteria in Equation 2 does not exceed 5% of the data set. However, these 5%—which mostly consists of accurate ground truth data—were used for model training. In Figure 1c, the ground truth data that meets the criteria in Equation 2 is shown (see the white solid line in Figure 1c), although it does not accurately represent the cusp region of the F2 layer trace. These incorrect ground truth data are also included in the training data set, though their proportion is expected to remain below a few percent, according to our observations.

3. Model and Training

Determining the layers traces h(f) from ionogram images requires solving a deep learning segmentation task. For this purpose, an effective U-net architecture, incorporating state-of-the-art attention mechanism, was implemented (see the block diagram in Figure 2). This architecture was originally developed by Oktay et al. (2018).

The training process follows the traditional two-step approach: the forward pass and the backward pass. During the forward pass, input data are passed through the network, producing an output, which is then compared to the ground truth to calculate the loss. In the backward pass, the model's weights are optimized by minimizing the loss function via backpropagation. Gradients are propagated from the loss function back through the network layers, with the chain rule applied at each step to compute gradients and iteratively update the weights. The Adam optimizer was used to minimize the loss function by optimizing the model's weights (Kingma & Ba, 2014):

$$W_t = W_{t-1} - \frac{\alpha m_t}{\sqrt{\nu_t} + \varepsilon} \tag{3}$$

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1)\frac{\partial L_t(W)}{\partial W} \tag{4}$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2)\left[\frac{\partial L_t(W)}{\partial W}\right]^2 \tag{5}$$

where L—loss function, t index—optimization step, W—weights of model, m—first moment estimate, v—second raw moment estimate, $\epsilon = 10^{-8}$ small constant for numerical stability, $\beta_1 = 0.9$, $\beta_2 = 0.999$–decay rates for the moment estimates, $\alpha = 0.0001$–learning rate, epoch—number of complete pass of entire training data set during the optimization process.

29935210, 2026, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JH001021, Wiley Online Library on [06/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

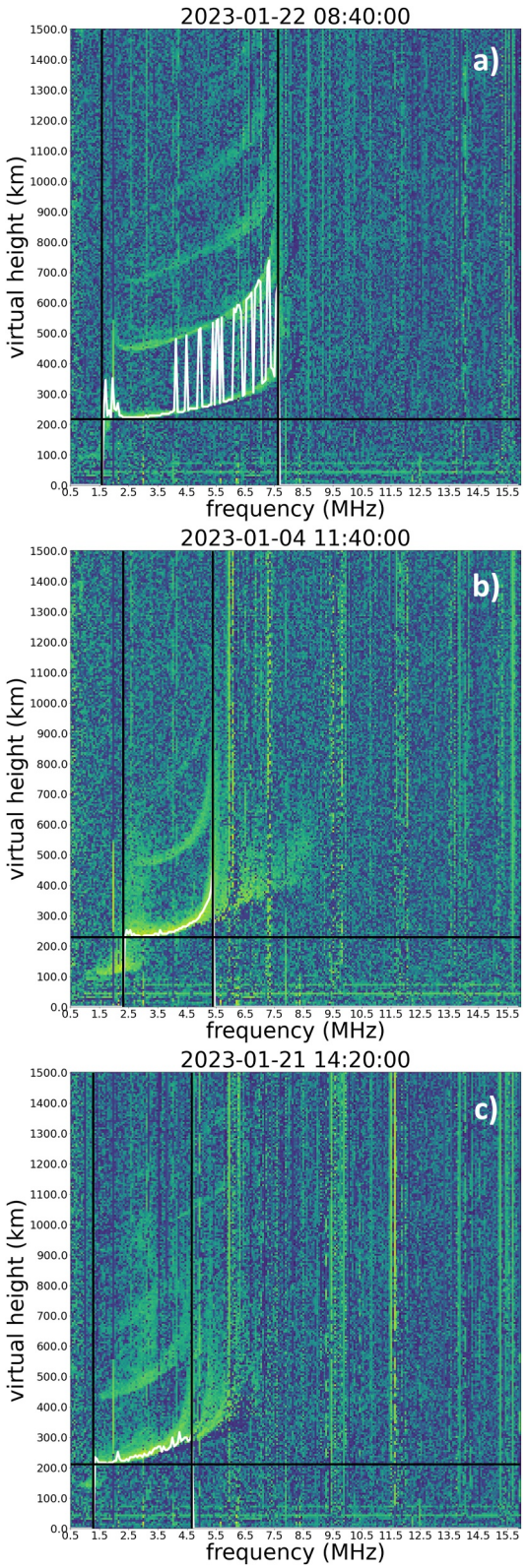


Figure 1.

The dice loss function was chosen for training due to the significant imbalance between pixels corresponding to layer traces and background pixels. Dice loss focuses on maximizing the overlap between the predicted mask and the ground truth, thus optimizing the model accordingly (see Equation 3).

$$L_{\text{Dice}} = 1 - \frac{2|X \cap Y|}{|X| + |Y|} \tag{6}$$

Where X—predicted positive pixels (mask), Y—ground truth positive pixels.

For model training, ionogram images were used as input, and ionogram masks determined using Equation 1 and the criteria defined in Equation 2 were used as ground truth. Layers with spread F typically have lower signal-to-noise ratios, and their patterns are less distinct than those without spread F. The use of such strict criteria for selecting ground truth data excludes a significant portion of ionograms containing spread F, and therefore the data set does not represent the full diversity of spread F patterns.

To address this limitation, a simple approach was implemented: input ionograms (see Figure 3a) were distorted by adding Gaussian noise (see Figure 3b), while the corresponding ground truth data remained unchanged (see Figure 3d). This method increases the difficulty of identifying layer traces while ensuring that the ground truth masks remain valid. At the same time, this approach prevents the model from focusing on amplitude and instead encourages it to learn the overall shape of the trace. We also note that preliminary tests without added Gaussian noise resulted in overfitting, manifested as small deviations in the predicted $\Delta h/\Delta f$ closely following the ground truth data. However, the data set collected using the argmax function and strict criteria for ground truth selection is significantly biased toward ionospheric traces with a high signal-to-noise ratio, which leads to poor performance when the signal-to-noise ratio is low. Thus, adding Gaussian noise to the training data is an important strategy, which mitigates the bias toward high signal-to-noise ratio data.

During the training process, the ground truth data were formed as binary masks giving the value 1 to h(f) ± trace width/2. In Figures 3a and 3d, the ionograms with and without added Gaussian noise are shown, together with the corresponding mask used as ground truth data.

4. Results

Three deep learning models with the same architecture (see the block diagram in Figure 2) were trained to separately identify E, F1, and F2 layer traces on ionogram images. The data set was divided into training and validation sets. For validation, data from 2021 were used. These data were completely excluded from the training data set, which otherwise spans the period from 2016 to 2024. Figure 4 shows the training and validation loss minimization over the course of the training epochs.

The reason the training loss is higher than the validation loss is that the training data are distorted and therefore more complex for the model to

Figure 1. (a) Ionogram from 2 January 2023 at 08:40 UT; (b) Ionogram from 1 April 2023 at 11:40 UT; (c) Ionogram from 21 January 2023 at 14:20 UT. Black lines indicate frequency and height bounds used to determine the positions of layer traces. White lines indicate the identified layer trace positions.

29935210, 2026, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JH001021, Wiley Online Library on [06/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

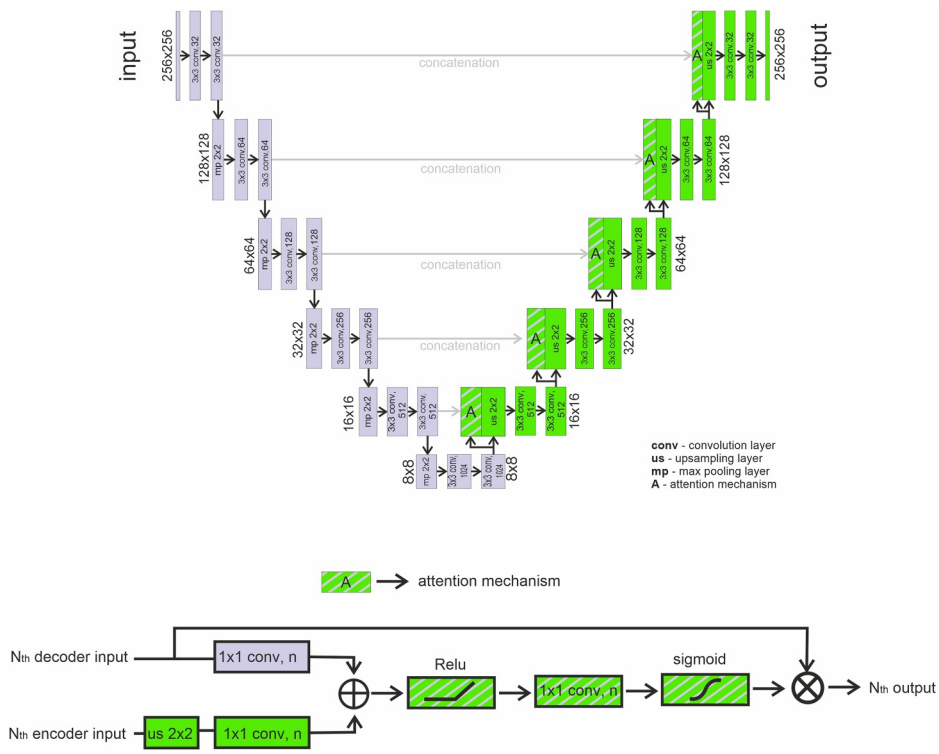


Figure 2. Block diagram of the U-Net model with attention mechanism.

recognize, compared to the undistorted validation data. In Figure 5, the dynamics of mean absolute errors (MAE) for f_{min}, f_{max}, and h_{min} are shown throughout the training process.

We don't rely solely on the minimum of the loss function, since our ground truth data are imperfect, as discussed in Section 2. Instead, we focus on the three aforementioned parameters, which can be reliably evaluated. The lowest MAE for determining f_{min}, f_{max}, and h_{min} (in pixels) achieved by deep-learning models are: E layer—0.79, 0.85, 0.54 pixels; F1 layer—0.66, 0.91, 0.58 pixels; F2 layer—1.19, 0.75, 1.24 pixels, respectively.

Figure 6 shows an ionogram at 8:40 UT on 22 January 2023, which was excluded from the training data set based on the criteria in Equation 2, but can now be accurately segmented for the F2 layer by the trained model. This demonstrates that, despite the data set being assembled using a simple argmax-based approach, the trained model makes decisions based on the shape of the layer trace rather than amplitude. As a result, the model can accurately recognize the F-layer trace in this case.

Figure 1c shows an example of a rare problematic case. A number of similar cases were not filtered out by the criteria in Equation 2. Typically, these ionograms exhibit a prominent spread F pattern near the cusp part of the layer, which in some cases makes it difficult even for a human expert to discern a correct trace. In Figure 7a, the result of segmentation by the trained model is shown. It can be seen that the model focused on the high-frequency cusp part of the spread F. However, according to URSI recommendations, it should instead focus on the low-frequency cusp part of the spread F.

It should be noted that this model behavior is not common. In general, the model handles similar patterns correctly. The mis-segmentation was likely caused by slight overfitting to a specific ionogram that was included in the training data set. Unfortunately, we cannot completely avoid such overfitting due to the imperfect nature of both the training and validation data sets, and there is no reliable way to analyze or detect it. Nonetheless, we consider it acceptable to achieve better MAE values for f_{min}, f_{max}, and h_{min} by training longer—even at the cost of rare mis-segmentations.

To address the mis-segmentation in cases of spread F, we implemented an additional data augmentation approach to create a new training data set. Each input ionogram was replicated and shifted along the frequency axis using

29935210, 2026, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JH001021, Wiley Online Library on [06/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

10.1029/2025JH001021

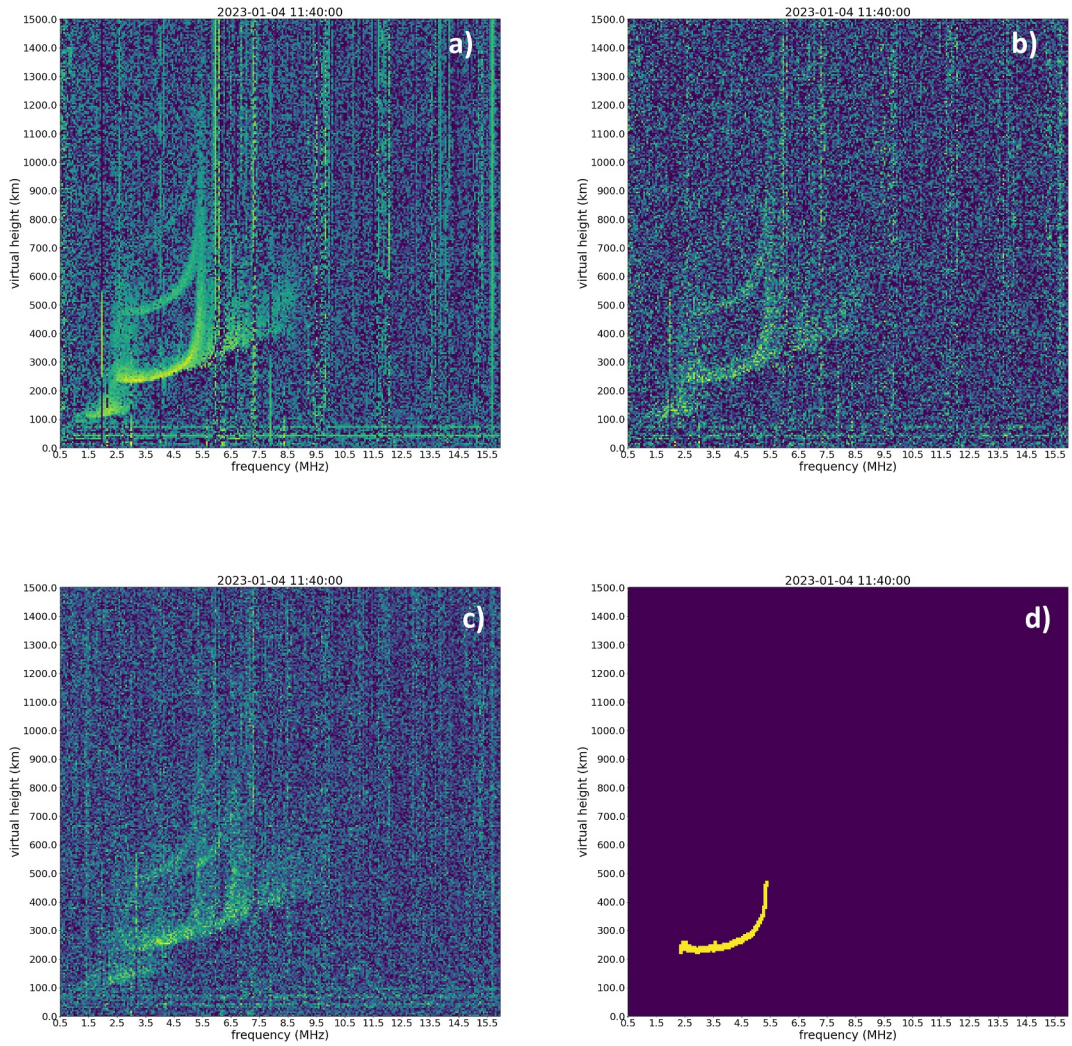


Figure 3. Ionogram from 4 January 2023 at 11:40 UT. (a) Original ionogram; (b) Ionogram distorted by Gaussian noise; (c) Composite of the original and shifted ionogram (using affine transformation), distorted by Gaussian noise; (d) Mask of the F2 layer trace (ground truth data).

affine transformation, then combined with the original (non-shifted) ionogram (see Figure 3c), while keeping the output/ground truth masks unchanged (Figure 3d). This simple technique distorts the original ionogram to imitate spread F patterns, while ensuring that the ground truth masks still correctly reflect the F-layer trace.

We trained a new model with the same architecture (see the block diagram in Figure 2) on the augmented data set. The segmentation result of this model for the ionogram at 14:20 UT on 21 January 2023 is shown in Figure 7b. The new model correctly focuses on the low-frequency part of the spread F. Although the model shows higher MAE values for $f_{\min} = 2.1$ pixels, $f_{\max} = 1.60$ pixels, and $h_{\min} = 1.83$ pixels compared to the main model, it successfully segments the F trace under spread F conditions in rare cases where the main model failed.

Since our ground truth data are imperfect, and only the parameters $f_{\min}$, $f_{\max}$, and $h_{\min}$ can be reliably evaluated (not the full trace segmentation), we visualize the model's performance on the same complex cases considered in Sherstyukov et al. (2024), thereby eliminating potential bias in selecting ionograms for model evaluation. These complex cases include:

1. Ionograms with low signal-to-noise ratio and poorly defined F1 and F2 traces (Figure 8a);
2. Presence of *Z*, *O*, and *X*-mode traces on the ionogram (Figure 8b);
3. Ionograms with additional cusps or hooks in the layer trace (Figure 8c);

29935210, 2026, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JH001021, Wiley Online Library on [06/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

10.1029/2025JH001021

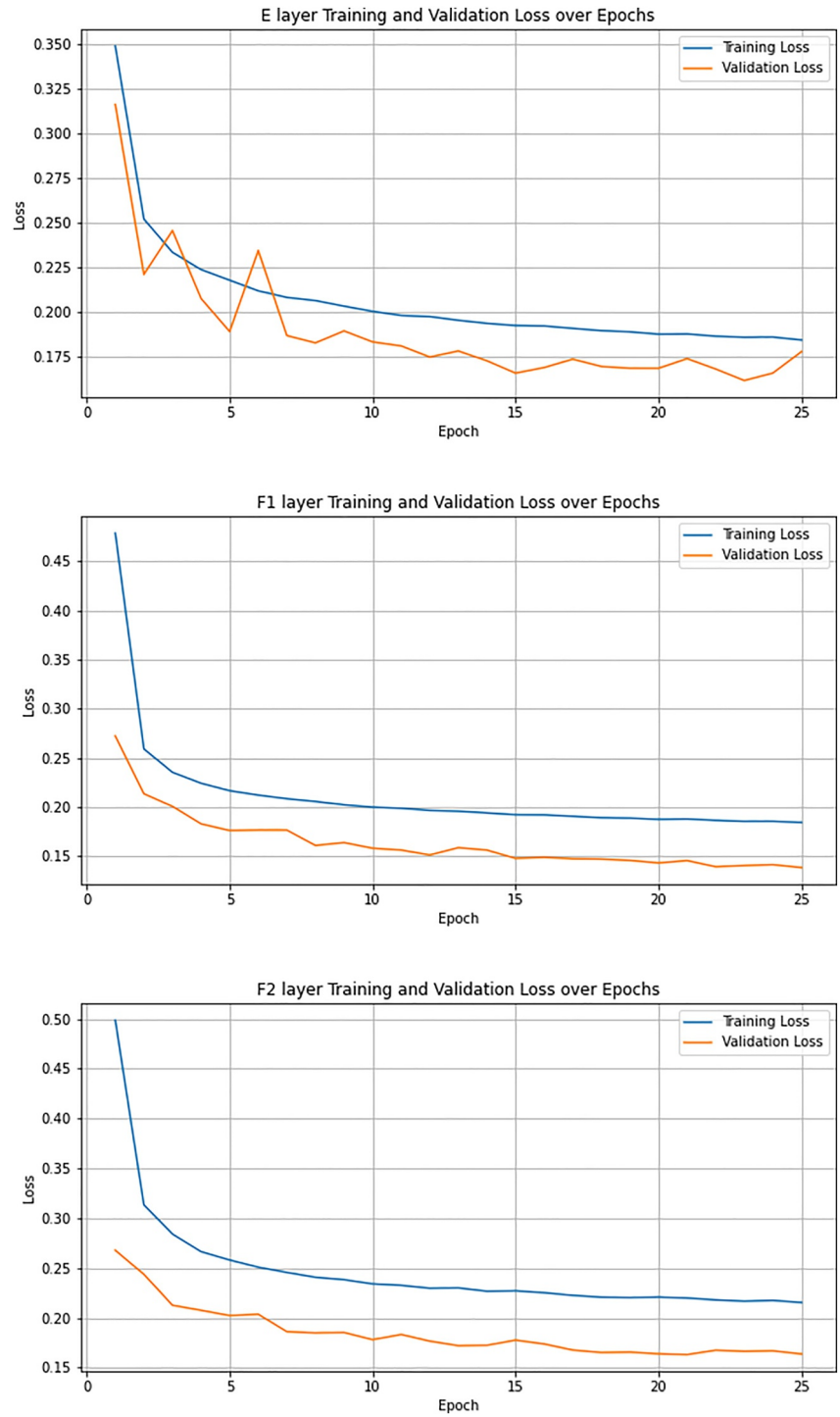


Figure 4. Training and validation loss values over 25 epochs for three separate models trained to segment E, F1, F2 layers.

4. Ionograms with frequency spread F (Figure 8d);
5. Ionograms with range spread F (Figure 8e);
6. Ionograms with blanketing Es layers and/or overlapping of Es multiple reflection traces and F-layer traces (Figure 8f).

29935210, 2026, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JH001021, Wiley Online Library on [06/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

10.1029/2025JH001021

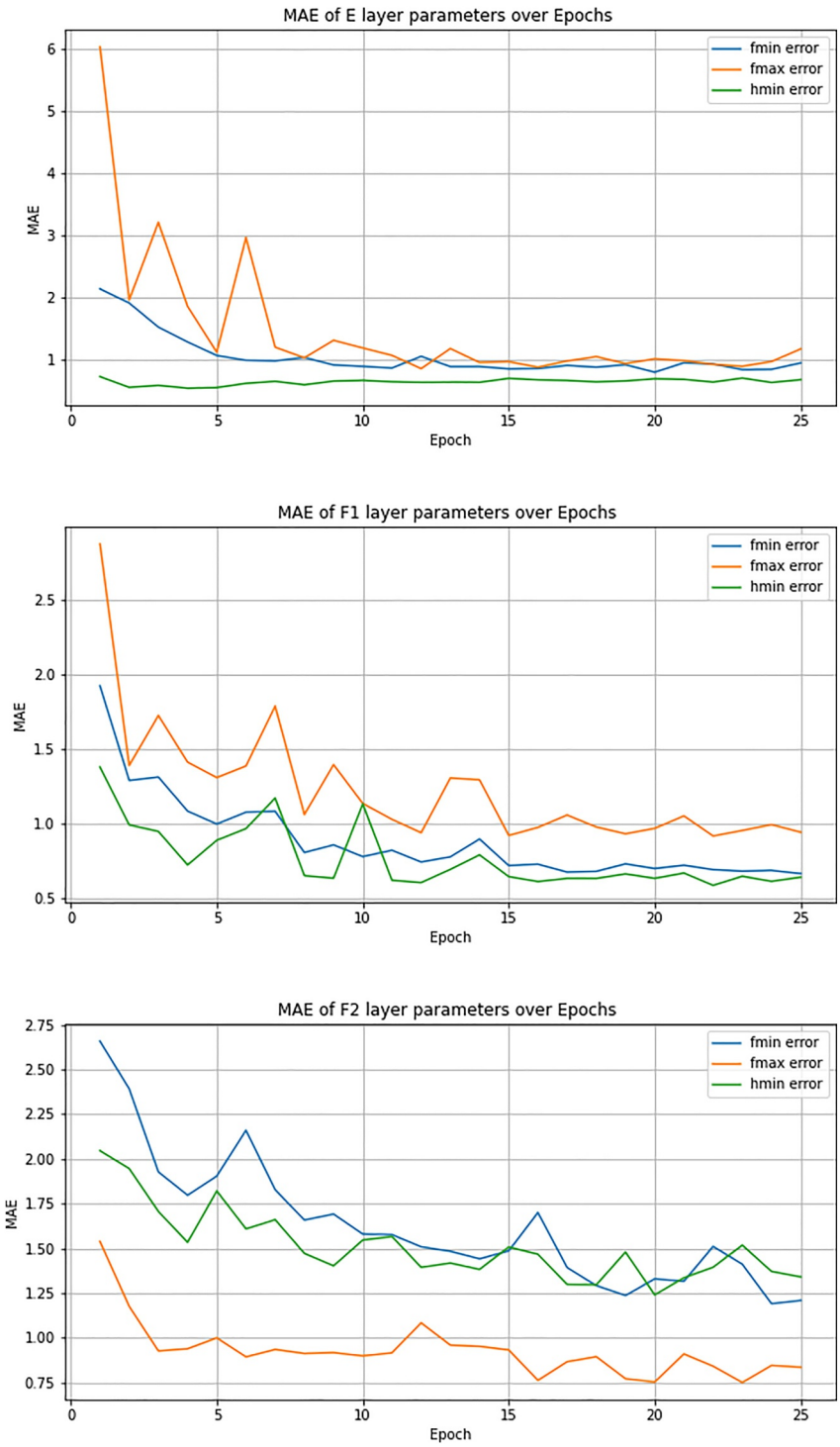


Figure 5. Mean absolute error (in pixels) on validation data over 25 epochs of training three separate models to determine $f_{\min}$, $f_{\max}$, and $h_{\min}$ for E, F1, F2 layers.

These cases are shown in Figure 8, where the masks (highlighted by the highest pixel intensities) produced by E, F1, F2 segmentation models are overlaid on each ionogram to illustrate the regions where the models are activated.

29935210, 2026, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JH001021, Wiley Online Library on [06/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

10.1029/2025JH001021

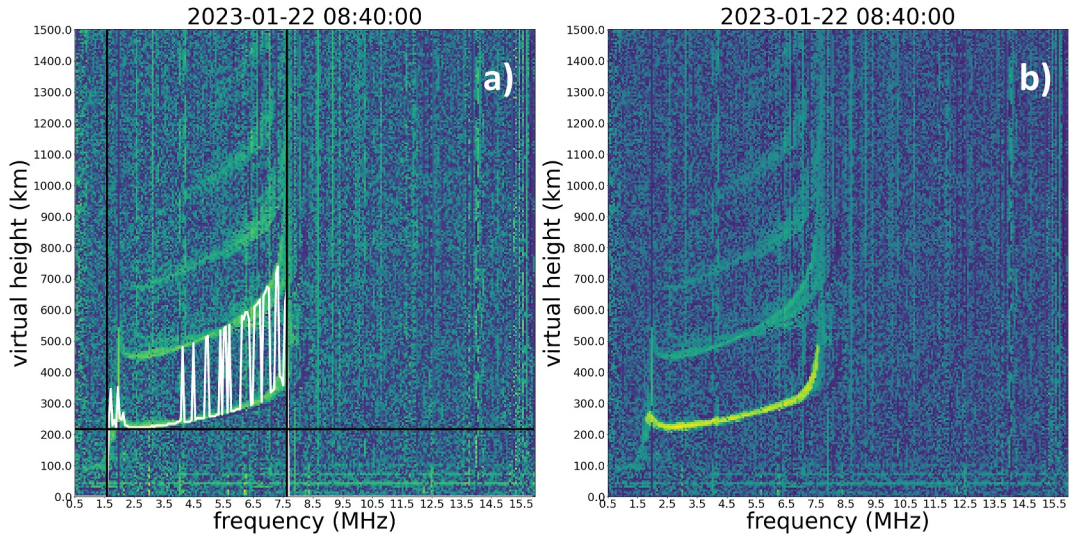


Figure 6. Ionogram from 22 January 2023 at 08:40 UT. (a) Position of the F2 layer trace determined using Equation 1. Black lines indicate the frequency and height bounds used to determine the positions of the layer trace. White lines indicate the identified layer trace positions; (b) Deep-learning model output mask indicating the F2 layer trace, displayed over the original ionogram. The mask is shown as increased pixel intensity on the ionogram image.

In general, the models correctly segment all layers in these ionograms (the masks for each layer are shown together in Figure 8). However, several minor segmentation inaccuracies can be noted:

Case 1 (Figure 8a): The low-frequency portion of the F2 layer with a U-shaped pattern (in the presence of the F1 layer) is not identified by the models, likely due to the very low signal-to-noise ratio in the ionogram. Nevertheless, this part can be visually distinguished upon close inspection.

Case 5 (Figure 8e): Although the model correctly ignores the additional trace from an oblique reflection—manifested as a secondary F2 trace above the main F2 layer trace (quasi-vertical reflection)—it does not fully segment the cusp of the F layer above 320 km altitude.

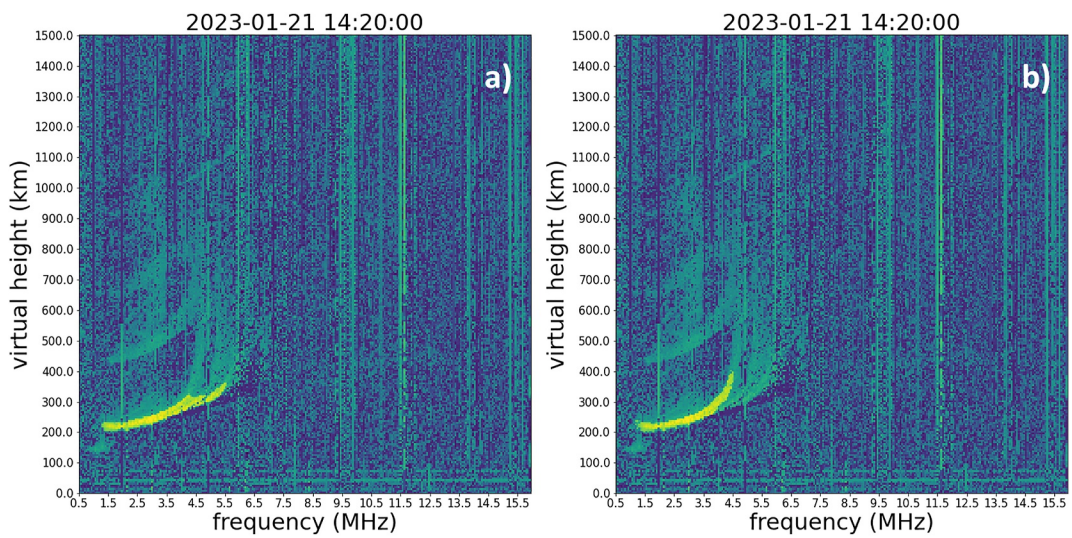


Figure 7. Ionogram from 21 January 2023 at 14:20 UT. (a) Output of deep-learning models trained to segment the F2 layer trace, shown as a mask overlaid on the original ionogram; (b) Output of deep-learning models trained to segment the spread F layer trace, also shown as a mask overlaid on the original ionogram. The mask is shown as increased pixel intensity on the ionogram image.

29935210, 2026, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JH001021, Wiley Online Library on [06/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

10.1029/2025JH001021

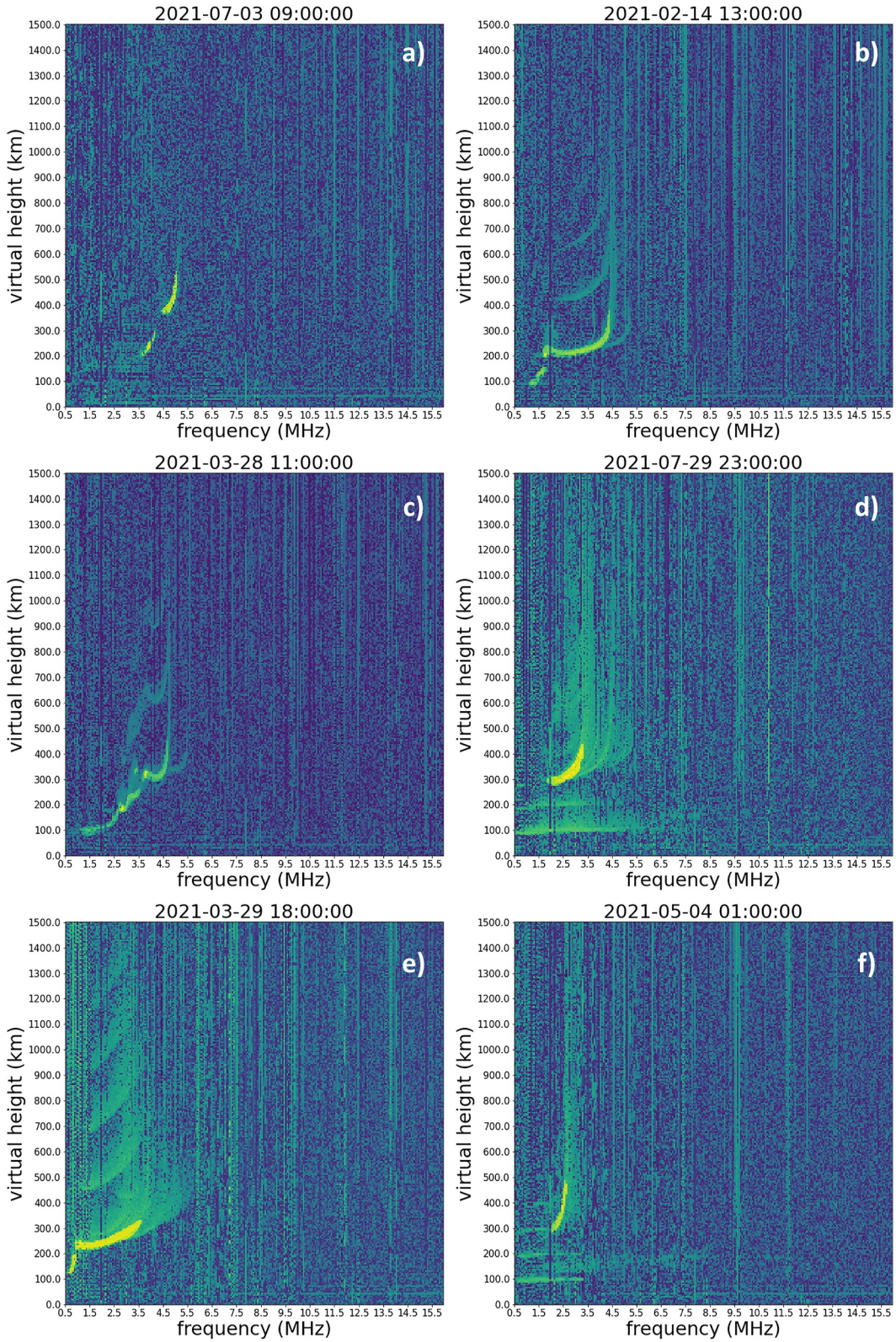


Figure 8. Challenging cases for automatic ionogram trace segmentation. (a) Ionogram with low signal-to-noise ratio and poorly defined F1 and F2 traces; (b) Presence of Z-mode, O-mode, and X-mode traces; (c) Additional cusps or hooks in the layer traces; (d) Ionogram with frequency spread F; (e) Ionogram with range spread F; (f) Ionogram with blanketing Es layers and/or overlapping of Es multiple reflection traces and F-layer traces. The outputs of deep-learning models trained to segment E, F1, and F2 layer traces are shown as masks overlaid on the original ionograms.

29935210, 2026, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JH001021, Wiley Online Library on [06/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

29935210, 2026, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JH001021, Wiley Online Library on [06/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Case 6 (Figure 8f): The models accurately identify the overlap between the F2-layer trace and the third-order Es-layer trace, avoiding false activation on the Es layer. However, it fails to detect the low-frequency portion of the F-layer trace observed below the third-order Es trace, at altitudes between 240 and 290 km.

5. Conclusions

The U-Net architecture with attention mechanism was used for pixel-wise segmentation of ionospheric layer traces on ionogram images. The uniqueness of the approach lies in the use of ground truth data assembled by applying a simple argmax function (see Equation 1) within height and frequency bounds. These bounds are defined based on a precise deep-learning regression model (see Sherstyukov, 2025) and additional criteria to exclude incorrect ground truth data (see Equation 2).

To reduce overfitting to amplitude values—an expected risk due to the use of the argmax function for assembling the training data set—Gaussian noise was added to the input ionograms during training. The imbalance of the assembled data set regarding the F-spread pattern was addressed by training an additional model using synthetic data. This data set was generated by replicating ionograms and shifting them along the frequency axis using affine transformations, then combining them with the original (non-shifted) ionograms while keeping the output/ground truth masks unchanged (see Figure 3).

This model demonstrated correct segmentation in rare cases featuring a spread-F pattern, where the base model tended to incorrectly identify the main trace of the F2 layer. However, its overall accuracy in estimating f_{min}, f_{max}, and h_{min} was lower than that of the base model not trained on synthetic F-spread cases. The base segmentation models showed the MAEs: for the E layer, f_{min} = 0.79 pixels (0.047 MHz), f_{max} = 0.85 pixels (0.051 MHz), and h_{min} = 0.54 pixels (3.10 km); for the F1 layer, f_{min} = 0.66 pixels (0.040 MHz), f_{max} = 0.91 pixels (0.055 MHz), and h_{min} = 0.58 pixels (3.33 km); and for the F2 layer, f_{min} = 1.19 pixels (0.071 MHz), f_{max} = 0.75 pixels (0.045 MHz), and h_{min} = 1.24 pixels (7.12 km).

Despite the high accuracy of deep learning models in segmenting entire ionospheric layer traces, rare and not typical trace patterns can still be incorrectly segmented. Visual inspection of the model's segmentation results on the validation data set revealed only one type of case affected by mis-segmentation. This case corresponds to the frequency F-spread type of the F2-layer trace with a low signal-to-noise ratio in the cusp region of the trace (see Figure 7). A strategy to mitigate such cases using synthetic ionogram data has been demonstrated. However, further verification of the deep learning models is required to determine whether other ionogram trace patterns prone to mis-segmentation exist.

The development of verification algorithms based on temporal variations of entire trace parameters, capable of identifying outliers and not typical trends, is also necessary. Such algorithms could be used to draw the operator's attention to cases of interest.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The ionograms, deep-learning models, and software used for model application and evaluation are available at Zenodo (Sherstyukov, 2025).

Acknowledgments
The research has been conducted with funding from the Research Council of Finland, Academy Research Fellowships, decision number: 371061. Open access publishing facilitated by Oulun yliopisto, as part of the Wiley - FinELib agreement.

References

Afraimovich, E. L., Astafyeva, E. I., Demyanov, V. V., Edemskiy, I. K., Gavrilyuk, N. S., Ishin, A. B., et al. (2013). A review of GPS/GLONASS studies of the ionospheric response to natural and anthropogenic processes and phenomena. *Journal of Space Weather and Space Climate*, *3*, A27. https://doi.org/10.1051/swsc/2013049

Akchurin, A., Smirnov, G., & Ildiryakov, V. (2021). Isolation of the small-scale and weak medium-scale TIDs on daytime midlatitude ionograms. In *2021 XXXIVth general assembly and scientific symposium of the international Union of radio science (URSI GASS)* (pp. 1–4). IEEE.

Galkin, I. A., & Reinisch, B. W. (2008). The new ARTIST 5 for all digisondes. *Ionosonde Network Advisory Group Bulletin*, *69*(8), 1–8.

Guo, L., & Xiong, J. (2023). Multi-scale attention-enhanced deep learning model for ionogram automatic scaling. *Radio Science*, *58*(3), 1–10. https://doi.org/10.1029/2022rs007566

Ivanov, V. A., Kurkin, V. I., Nosov, V. E., Uryadov, V. P., & Shumaev, V. V. (2003). Chirp ionosonde and its application in the ionospheric research. *Radiophysics and Quantum Electronics*, *46*(11), 821–851. https://doi.org/10.1023/b:raqe.0000028576.51983.9c

Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*.

Kozlovsky, A., Stober, G., Sherstyukov, R., Lester, M., Belova, E., Kero, J., et al. (2025). Atmospheric gravity waves and medium Scale traveling ionospheric disturbances at auroral latitudes. *Surveys in Geophysics*, 1–17. https://doi.org/10.1007/s10712-025-09880-0

Kozlovsky, A., Turunen, T., & Ulich, T. (2013). Rapid-run ionosonde observations of traveling ionospheric disturbances in the auroral ionosphere. *Journal of Geophysical Research: Space Physics*, *118*(8), 5265–5276. https://doi.org/10.1002/jgra.50474

Kozlovsky, A., Shalimov, S., Kero, J., Raita, T., & Lester, M. (2018). Multi-instrumental observations of nonunderdense meteor trails. *Journal of Geophysical Research: Space Physics*, *123*(7), 5974–5989. https://doi.org/10.1029/2018ja025405

Kvammen, A., Vierinen, J., Huyghebaert, D., Rexer, T., Spicher, A., Gustavsson, B., & Floberg, J. (2024). NOIRE-Net – A convolutional neural network for automatic classification and scaling of high-latitude ionograms. *Frontiers in Astronomy and Space Sciences*, *11*, 1289840. https://doi.org/10.3389/fspas.2024.1289840

Maruyama, T., Kato, H., & Nakamura, M. (2003). Ionospheric effects of the Leonid meteor shower in November 2001 as observed by rapid run ionosondes. *Journal of Geophysical Research*, *108*(A8). https://doi.org/10.1029/2003ja009831

Mochalov, V., & Mochalova, A. (2019). Application of deep learning to recognize ionograms. *Russian Open Conference on Radio Wave Propagation (RWP)*, *1*, 477–479.

Moges, S. T., Kozlovsky, A., Sherstyukov, R. O., & Ulich, T. (2024). Solar activity dependence of traveling Ionospheric disturbance amplitudes using a rapid-run ionosonde in high latitudes. *Journal of Geophysical Research: Space Physics*, *129*(10), e2024JA033013. https://doi.org/10.1029/2024ja033013

Moges, S. T., Sherstyukov, R. O., Kozlovsky, A., Ulich, T., & Lester, M. (2024). Statistics of traveling ionospheric disturbances at high latitudes using a rapid-run ionosonde. *Journal of Geophysical Research: Space Physics*, *129*(4), e2023JA031694. https://doi.org/10.1029/2023ja031694

Norberg, J., Vierinen, J., Roininen, L., Orispää, M., Kauristie, K., Rideout, W. C., et al. (2018). Gaussian Markov random field priors in ionospheric 3-D multi-instrument tomography. *IEEE Transactions on Geoscience and Remote Sensing*, *56*(12), 7009–7021. https://doi.org/10.1109/tgrs.2018.2847026

Oktay, O., Schlemper, J., Folgoc, L. L., Lee, M., Heinrich, M., Misawa, K., & Rueckert, D. (2018). Attention u-net: Learning where to look for the pancreas. *arXiv preprint arXiv:1804.03999*.

Pezzopane, M., & Scotto, C. (2005). The INGV software for the automatic scaling of foF2 and MUF(3000)F2 from ionograms: A performance comparison with ARTIST 4.01 from Rome data. *Journal of Atmospheric and Solar-Terrestrial Physics*, *67*(12), 1063–1073. https://doi.org/10.1016/j.jastp.2005.02.022

Pezzopane, M., & Scotto, C. (2007). Automatic scaling of critical frequency foF2 and MUF(3000)F2: A comparison between Autoscala and ARTIST 4.5 on Rome data. *Radio Science*, *42*(4), 1–17. https://doi.org/10.1029/2006rs003581

Pezzopane, M., & Scotto, C. (2008). A method for automatic scaling of F1 critical frequencies from ionograms. *Radio Science*, *43*(2), 1–12. https://doi.org/10.1029/2007rs003723

Reinisch, B. W., Galkin, I. A., Khmyrov, G., Kozlov, A., & Kitrosser, D. (2005). Automated collection and dissemination of ionospheric data from the digisonde network. *Advances in Radio Science*, *2*, 241–247. https://doi.org/10.5194/ars-2-241-2004

Scotto, C., & Pezzopane, M. (2007). A method for automatic scaling of sporadic E layers from ionograms. *Radio Science*, *42*(2), 1–5. https://doi.org/10.1029/2006rs003461

Sherstyukov, R. (2025). Synthetic-Data-Driven supervised learning for pixel-wise recognition of ionogram layer traces [Dataset and computer software]. *Zenodo*. https://doi.org/10.5281/zenodo.17143326

Sherstyukov, R., Kozlovsky, A., Ulich, T., & Moges, S. (2023). Deep learning for ionogram parameters scaling at polar region ionosphere. In *2023 XXXVTH general assembly and scientific symposium of the international Union of radio science (URSI GASS)* (pp. 1–4). IEEE.

Sherstyukov, R., Moges, S., Kozlovsky, A., & Ulich, T. (2024). A deep learning approach for automatic ionogram parameters recognition with convolutional neural networks. *Earth and Space Science*, *11*(10), e2023EA003446. https://doi.org/10.1029/2023ea003446

Sherstyukov, R. O., Akchurin, A. D., & Sherstyukov, O. N. (2018). Collocated ionosonde and dense GPS/GLONASS network measurements of midlatitude MSTIDs. *Advances in Space Research*, *61*(7), 1717–1725. https://doi.org/10.1016/j.asr.2017.11.026

Xiao, Z., Wang, J., Li, J., Zhao, B., Hu, L., & Liu, L. (2020). Deep-learning for ionogram automatic scaling. *Advances in Space Research*, *66*(4), 942–950. https://doi.org/10.1016/j.asr.2020.05.009

Yusupov, K., & Akchurin, A. (2014). Meteor induced layers in 2013 observed by ionosonde with high cadence. In *2014 XXXIth URSI general assembly and scientific symposium (URSI GASS)* (pp. 1–4). IEEE.

29935210, 2026, 3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JH001021, Wiley Online Library on [06/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License